

RAJA KUMAR

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SUMMARY

Final-year B.Tech CSE student (CGPA 9.11) specializing in AI/ML, Generative AI, and intelligent systems. Experienced in building RAG applications, AI agents & vector databases, using Python, FastAPI, Docker, and CI/CD.

SKILLS

Programming Languages: Java, Python, C, SQL

Frameworks & Libraries: FastAPI, Flask, LangChain, LangGraph, Next.js, React, Express, TensorFlow

Tools & Platforms: AWS, Docker, Git, GitHub Actions, VS Code, PyCharm, Jupyter Notebook, CI/CD

AI/ML & Software Engineering: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, System Design

Core Competencies: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

AI-Map: Local RAG & Vector Database Platform | Python, HNSW, Ollama, Docker, GitHub Actions GitHub

- Built a Vector Database from scratch using HNSW ($O(\log N)$), KD-Tree, and Brute Force algorithms in Python.
- Implemented an offline RAG pipeline with Ollama for document retrieval and AI-powered question answering.
- Designed an interactive web interface with PCA visualization and benchmarking across multiple search algorithms and distance metrics.
- Containerized the application with Docker Compose and automated testing and Docker builds using GitHub Actions CI/CD.

Quant Agent: Multi-Agent Trading System | Python, LangGraph, LangChain, Flask GitHub

- Developed a multi-agent AI trading platform using LangGraph for autonomous market analysis and trade decision orchestration.
- Integrated GPT-4o, LLaVA, and TA-Lib indicators (RSI, MACD), reducing false trading signals by 30%.
- Engineered scalable Flask REST APIs for real-time inference, agent communication, and consensus-based trade recommendations.
- Automated containerized deployments using Docker, GitHub Actions, and CI/CD pipelines.

MIRAI: Autonomous AI Combat System | Python, FastAPI, FAISS, Ollama, Next.js GitHub

- Architected an AI backend inspired by NVIDIA ACE using FAISS vector retrieval, behavioral profiling, PyTorch, and XGBoost ensemble inference for adaptive boss intelligence.
- Designed an online reinforcement learning-inspired feedback loop with telemetry-driven intent prediction, episodic memory, and adaptive strategy selection.
- Built a FastAPI-based Cognitive OS using Ollama (Llama 3.2) for explainable AI reasoning and real-time tactical decision-making.

EDUCATION

KIIT UNIVERSITY

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 9.11

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Discrete Mathematics, Machine Learning, Data Mining

OTHER EXPERIENCE

SwasthyaPro

Jul 2025 – Sep 2025

Digital Marketing Intern (Remote)

- Improved organic search visibility and analyzed 1000+ monthly traffic insights using Google Analytics and SEMrush to optimize SEO performance.

ACHIEVEMENTS & CERTIFICATIONS

- Completed [AWS Cloud Architecting](#), [AICTE Data Science Internship \(Siemens\)](#), and [Data Science, AI/ML & GenAI](#).
- Solved 300+ DSA problems and secured 1st Runner-Up at KIIT Hackathon [UiPath Resume Screener](#).